

Nawin Viswajith B

LinkedIn: [linkedin.com/in/nawinviswajith](https://www.linkedin.com/in/nawinviswajith)

GitHub: github.com/nawin-viswajith

Email: nawinviswajith@gmail.com

Mobile: +91 86083 43975

Professional Summary

AI & Machine Learning Engineer with hands-on production experience building enterprise AI systems, Machine Learning pipelines, and Generative AI applications. Experienced in Python, FastAPI, scikit-learn, PyTorch, LangGraph, Azure OpenAI, and ChromaDB, with expertise in Retrieval-Augmented Generation (RAG), multi-agent orchestration, LLM evaluation, and scalable backend AI services. Delivered 5+ production AI systems at Prodapt, achieving an 92.1% RAGAS quality score and a 94.4% reduction in vector database memory usage. Backed by a granted patent and two peer-reviewed publications.

Technical Skills

- **Languages & Core ML:** Python, SQL, NumPy, Pandas, scikit-learn
- **Generative AI and LLMs:** LangGraph, LangChain, DSPy, Large Language Models(LLMs), Retrieval-Augmented Generation (RAG), RAGAS, Agentic Workflows, Multi-Agent Systems, LLM-as-a-judge, Prompt Engineering, HuggingFace, Text Processing & Chunking, Vector Embeddings, LLM Evaluation, Azure OpenAI, Google Gemini, Contrast-RAG, Reflection Loops, AI Safety, Guardrails, LLMops, MLOps
- **Backend & Cloud APIs:** FastAPI, REST APIs, Azure Cloud Services, SQLAlchemy, Redis, JSON, Postman
- **Databases and Vector Stores:** MySQL, SQLite, ChromaDB, FAISS
- **DevOps and Security:** Git, Azure, ECDH-P256, AES-256-GCM, Presidio (PII detection)
- **Computer Vision and Tools:** OpenCV, Tree-Sitter, Tesseract OCR, EasyOCR, CLIP, openpyxl, MediaPipe

Experience

Prodapt

Chennai, India

Overall Tenure: July 2025 – Present

AI & Machine Learning Engineer

November 2025 – Present

- **Assessment Engine and Multi-Agent Question Generation:** Designed and deployed 5 core FastAPI microservices integrated with Azure OpenAI services within a 20+ microservice enterprise AI assessment platform – multi-agent question generation engine, candidate report service, AST code parser, bulk text chunking and upload pipeline, and image guardrail service – reducing per-question cost to \$0.0042, achieving 89.5% RAGAS quality score (1,000-question LLM-as-judge test run), and cutting vector DB RAM by 94.4% (9.22 GB to 508.9 MB) via LRU-partitioned ChromaDB sharding; awarded Rising Rookie of the Month.
- **Enterprise AI Data Gateway:** Architecting a privacy-first, local-LLM-driven data workflow over 17 schemas and 530+ tables (millions of records), featuring multi-agent correction loops for complex SQL and PowerBI DAX generation. Implementing strict RBAC for multi-tier executive access, a Knowledge Graph with metadata filtering for optimized sub-schema retrieval, and semantic caching via data snapshots to minimize database queries, scoping an initial 85% text-to-query accuracy target.
- **Document Intelligence Pipeline:** Built a dual-agent document analysis and text chunking pipeline integrated with Azure services and sub-ms SharePoint search (Path Trie + Trigram Index) via a 5-tier skill-matching cascade (exact match, fuzzy, semantic cache, vector search, LLM batch judge); 3-layer PII detection (Presidio ML + Aadhaar/PAN/passport/IP regex), greater than 98% success rate via LLM fallback chains, and 15+ Prometheus metrics.
- **Attrition Prediction System:** Architecting an end-to-end employee attrition prediction pipeline using scikit-learn and Pandas (ETL and Feature Store phase, active development) targeting 6-month exit forecasting via dual-model routing – tenured model (appraisal, compensation, promotion) and new-joiner model (engagement, onboarding feedback) – backed by a temporal Feature Store across 25+ domain tables and macroeconomic signals (India GDP, IT growth, Naukri, Gartner).

Graduate Trainee

July 2025 – November 2025

- Completed full-stack AI development training (Python, FastAPI, React, MySQL, SQLAlchemy); built an AI-powered network tower load monitoring system and an end-to-end RFM-based dynamic dunning platform to optimise revenue recovery.

- Selected among 100 students nationwide for the Ministry of Defence i4D internship; developed real-time computer vision solutions (CNN, YOLO) for PPE wear detection, PCB fault detection, and body movement tracking for VR safety training.

Projects

Trunk (On-Device AI & Multi-Agent Playground)

Website | GitHub

Kotlin · Android SDK · Llama.cpp / MLC LLM · AI · SQLite

- Engineered a privacy-first Android application for on-device LLM inference, enabling completely offline, cloud-free AI interactions without backend dependencies.
- Designed and implemented a visual, no-code “Playground” allowing users to seamlessly chain multiple AI agents (e.g., drafter, refiner, formatter) into automated pipelines directly on mobile hardware.

Multi-Agent Professional Assistant (LangGraph)

Hackathon, Prodapt

Python · FastAPI · LangGraph · LangChain · Azure OpenAI · ChromaDB · MySQL · React · JWT

- Architected a 7-agent LangGraph StateGraph pipeline (Router → Specialist → Humanizer) with dual RAG pipelines (all-MiniLM-L6-v2 at ~80 ms for personal notes; Azure OpenAI embeddings for 59 company policy PDFs), priority-based keyword routing with LLM fallback, per-user ChromaDB collections, LRU memory eviction (24 h TTL), and a React frontend with employee directory search across 1,500+ records.

Enterprise Agentic Text-to-SQL Platform (Reflection + MCP)

GitHub

Python · FastAPI · LangGraph · Azure OpenAI · ChromaDB · Sentence-Transformers · MySQL · React · JWT

- LangGraph Reflection workflow with schema-aware Retrieval-Augmented Generation (ChromaDB/MMR), automated RAGAS evaluation, strict SQL governance (SELECT-only, EXPLAIN pre-checks, auto-retry on schema mismatch), MCP-compatible API surface for external agent tool usage, and React dashboard with LangSmith telemetry traces.

Patents, Publications and Certifications

Patents:

- Virtual Reality Powered Flight Training, Application No. 202441083983 A November 2024

Journal:

- Flight Procedure Trainer using VR, TIJER May 2024
- AI Based Collision Avoidance System for Unmanned Vessels, TIJER June 2023

Certifications:

- Oracle OCI 2025: Generative AI Professional, AI Vector Search, Data Science Professional October 2025

Education

Bachelor of Technology in Artificial Intelligence and Data Science | CGPA: 8.13

2021 – 2025

Sri Sairam Engineering College, Chennai

- Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, System Design and Architecture, Data Structures and Algorithms, Database Management System

Honours

- Rising Rookie of the Month (Prodapt):** Awarded for architecting a production-grade Agentic RAG pipeline with ReAct loops and VectorDB partitioning.
- iDEX – Innovate4Defence:** Selected among 100 students nationwide for the Ministry of Defence iDEX i4D internship programme.
- Top OnePlus Beta Tester:** Handpicked for OxygenOS 12 Closed beta testing; Top Tester with 1,983+ bugs reported.